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- Collaboration is encouraged, but all write-ups must be done individually and independently, and mention your collaborator(s) if any. Same rules apply for codes written for any programming assignments (i.e., write your own code; we will run plagiarism checks on codes).
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1. (4 points) [APPLYING D-SEP PRECISELY!] Let's clarify some implications of d-sep sketched in class. Let X, Y be two **non-adjacent** r.v.s in a BN G , and let X, Y have at least two common ancestors and at least two common descendants. Note that X and Y can either be in an ancestor-descendant relation (denoted *adr*), or not (denoted *non-adr*). Answer if each statement below is True/False.

(a) (1 point) For *adr* case, $(X \perp Y \mid \text{all common ancs.})$ holds (i.e., this CI is in $I(G)$ for any G).

Solution: False

(b) (1 point) For *adr* case, $(X \perp Y \mid \text{Pa}_X \cup \text{Pa}_Y)$ doesn't hold in some G .

Solution: False

(c) (1 point) For *non-adr* case, whether $(X \perp Y \mid \text{a common anc. } A)$ holds (or not) can depend on which common anc. is chosen as A .

Solution: True

(d) (1 point) For *non-adr* case, whether $(X \perp Y \mid \text{a common desc. } D)$ holds (or not) can depend on which common desc. is chosen as D .

Solution: False

2. (5 points) [ISING MEETS BOLTZMANN] Consider a MN H over n nodes (such as a grid graph, with $i \sim j$ indicating edges (X_i, X_j) in H). Ising model defines a joint distbn. over ± 1 -valued r.v.s X as $P(x) = \frac{1}{Z'} \exp \left\{ - \sum_{i \sim j} a_{ij} x_i x_j - \sum_i b_i x_i \right\}$, whereas Boltzmann machine defines a joint distbn. over 0/1-valued r.v.s. using the same form as $P(x) = \frac{1}{Z} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}$, with the result being a single non-zero contribution of each edge (X_i, X_j) 's potential when $X_i = X_j = 1$.

(a) (2 points) Show how any Ising model can be reformulated as a Boltzmann machine (with -1 mapped to 0) by showing how the node/edge potentials and partition function of both models are related.

Solution: Solution: We use the transformation $x_B = \frac{x_I + 1}{2}$, where x_B represents 0/1 valued r.v.s and x_I represents ± 1 valued r.v.s.

Then,

$$P_I(X) = \frac{1}{Z'} \exp \left\{ - \sum_{i \sim j} a_{ij} x_{I_i} x_{I_j} - \sum_i b_i x_i \right\}$$

$$= \frac{1}{Z'} \exp \left\{ - \sum_{i \sim j} a_{ij} (2x_{B_i} - 1)(2x_{B_j} - 1) - \sum_i b_i 2x_{B_i} \right\}$$

$$= \frac{1}{Z'} \exp \left\{ - \sum_{i \sim j} a_{ij} (4x_{B_i} x_{B_j} - 2x_{B_i} - 2x_{B_j} + 1) - \sum_i b_i 2x_{B_i} \right\}$$

$$= \frac{1}{Z'} \exp \left\{ - \sum_{i \sim j} (a_{ij} 4x_{B_i} x_{B_j} - a_{ij} 2x_{B_i} - a_{ij} 2x_{B_j} + a_{ij}) - \sum_i b_i 2x_{B_i} \right\}$$

$$= \frac{1}{Z''} \exp \left\{ - \sum_{i \sim j} a_{ij} 4x_{B_i} x_{B_j} - \sum_j \left(\sum_i a_{ij} \delta_{ij} \right) 2x_{B_i} / 2 - \sum_j \left(\sum_i a_{ij} \delta_{ij} \right) 2x_{B_j} / 2 - \sum_i b_i 2x_{B_i} \right\}$$

where, $\frac{1}{Z''} = \frac{1}{Z'} \exp \left\{ \sum_{i \sim j} (-a_{ij}) \right\}$, $\delta_{ij} = 1$ if there is an edge between i and j and 0 otherwise.

The factor of 2 is introduced since every factor is counted twice in $\sum_j \sum_i$ whereas it was counted only once in $\sum_{i \sim j}$.

Now, by symmetry,

$$\sum_j \left(\sum_i a_{ij} \delta_{ij} \right) 2x_{B_j} = \sum_i \left(\sum_j a_{ji} \delta_{ji} \right) 2x_{B_i}$$

But, a_{ij} and δ_{ij} are both symmetric, i.e. $a_{ij} = a_{ji}$

$$\implies \sum_j \left(\sum_i a_{ij} \delta_{ij} \right) 2x_{B_j} = \sum_i \left(\sum_j a_{ij} \delta_{ij} \right) 2x_{B_i}$$

$$\therefore P_I(X) = \frac{1}{Z''} \exp \left\{ - \sum_{i \sim j} a_{ij} 4x_{B_i} x_{B_j} - 2 \sum_i \left(\sum_j a_{ij} \delta_{ij} + b_i \right) x_{B_i} \right\}$$

Therefore,

$$a_{ij} = \frac{c_{ij}}{4}, b_i = \frac{d_i}{2} - \sum_j a_{ij} \delta_{ij}$$

- (b) (2 points) Derive the conditional distribution $P(X_i | \text{Nbrs}_{X_i})$ for the Boltzmann machine, and express it as a “neuron activation model” (i.e., in terms of the function $\text{sigmoid}(z) = e^z / (1 + e^z)$).

Solution: Solution: $P(X_i | \text{Nbrs}_{X_i}) = \frac{P(X_i, \text{Nbrs}_{X_i})}{P(\text{Nbrs}_{X_i})}$

$$\begin{aligned}
 &= \frac{\sum_{X \setminus (X_i \cup \text{Nbrs}_{X_i})} P(X_i, \text{Nbrs}_{X_i})}{\sum_{X \setminus (\text{Nbrs}_{X_i})} P(\text{Nbrs}_{X_i})} \\
 &= \frac{\sum_{X \setminus (X_i \cup \text{Nbrs}_{X_i})} P(X)}{\sum_{X \setminus (\text{Nbrs}_{X_i})} P(X)} \\
 &= \frac{\sum_{X \setminus (X_i \cup \text{Nbrs}_{X_i})} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}}{\sum_{X \setminus (\text{Nbrs}_{X_i})} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}} \\
 &= \frac{\sum_{X \setminus (X_i \cup \text{Nbrs}_{X_i})} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}}{\sum_{X \setminus (X_i \cup \text{Nbrs}_{X_i})} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\} + \sum_{X_i} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}}
 \end{aligned}$$

Dividing by $\sum_{X_i} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}$,

$$\Rightarrow \frac{\frac{\sum_{X \setminus (X_i \cup \text{Nbrs}_{X_i})} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}}{\sum_{X_i} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}}}{\frac{\sum_{X \setminus (X_i \cup \text{Nbrs}_{X_i})} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}}{\sum_{X_i} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}} + 1}$$

Let $z = \log \left(\frac{\sum_{X \setminus (X_i \cup \text{Nbrs}_{X_i})} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}}{\sum_{X_i} \exp \left\{ - \sum_{i \sim j} c_{ij} x_i x_j - \sum_i d_i x_i \right\}} \right)$

Then, $P(X_i | \text{Nbrs}_{X_i}) = \frac{e^z}{1 + e^z}$

- (c) (1 point) Is Ising model in the exponential family of distributions? Explain using the defn. of exponential family seen in the last assignment.

Solution: Solution: An exponential distribution is given by:

$$p(x|\theta) = h(x) \exp \{ \eta(\theta)^\top t(x) - a(\theta) \}$$

Consider, the Ising model in matrix form

$$P(X | A, B) = \frac{1}{Z} \exp\{X^T A X - B^T X\}$$

Consider, matrices P and L s.t. $A = L^T L$, and $B^T = -2P^T L$

$$(LX + P)^T (LX + P) = X^T L^T L X + X^T L^T P + P^T L X + P^T P$$

$$= (LX + P)^T (LX + P) = X^T L^T L X + 2P^T L X + P^T P$$

Comparing with $X^T A X - B^T X$, we have,

$$P(X | A, B) = \frac{1}{Z} \exp\{(LX + P)^T (LX + P) - P^T P\}, \text{ where,}$$

$$A = L^T L, \text{ and } B^T = -2P^T L$$

Thus, the Ising model is in the exponential family, with $\eta(\theta) = I_{p \times p}$, $t(x) = (LX + P)^T (LX + P)$, and $a(\theta) = P^T P$

3. (7 points) [(OVER)-PARAMETERIZATIONS]

- (a) (2 points) Consider a complete graph MN over 3 binary r.v.s parameterized either as a single clique potential or as a product of 3 pairwise potentials. The former requires 8 parameters to write down the clique potential table, whereas the latter requires 12 parameters to write down the pairwise potentials. Can you explain how to resolve the apparent contradiction between these two different parameter counts?

Solution:

The pairwise potentials only capture interactions between two variables, and do not capture interactions between higher order factors. For any n , pairwise parametrization will require $4 \times \binom{n}{2}$ parameters whereas the parametrization over the whole clique will require 2^n parameters (for binary r.v.s).

Intuitively, for a small n , the number of higher order factors are lower, and the local potentials have more than enough information to capture the joint distribution. However, as n increases, the number of higher order interactions increases exponentially and local potentials no longer have enough information to completely represent the joint distribution.

Note that for small values of n , there are more factors in the pairwise parametrization than required for representing the joint distribution. This is explained by the fact that though there are more parameters, these have limited degrees of freedom. For example, while multiplying factors to get the joint distribution, we have to multiply them in a way that the common part "matches up" as shown in the table below, i.e., not all configurations of factors are allowed in the joint distribution. This explains the apparent contradiction in the number of parameter counts.

In the table below, $\mathbb{P}'$ denotes unnormalized probability and α_{ab} , β_{bc} and γ_{ca} are the pairwise factors. Note how only certain configurations of triplets of α_{ab} , β_{bc} , and γ_{ca} are present in the joint distribution.

	$\mathbb{P}'_1(A, B, C)$	$\mathbb{P}'_2(A, B, C)$
$A, B, C = 0, 0, 0$	$\alpha_{1ab}\beta_{1bc}\gamma_{1ca}$	θ_{1abc}
$A, B, C = 0, 1, 0$	$\alpha_{2ab}\beta_{3bc}\gamma_{1ca}$	θ_{2abc}
$A, B, C = 1, 0, 0$	$\alpha_{3ab}\beta_{1bc}\gamma_{2ca}$	θ_{3abc}
$A, B, C = 1, 1, 0$	$\alpha_{4ab}\beta_{3bc}\gamma_{2ca}$	θ_{4abc}
$A, B, C = 1, 0, 1$	$\alpha_{3ab}\beta_{2bc}\gamma_{4ca}$	θ_{5abc}
$A, B, C = 1, 1, 1$	$\alpha_{4ab}\beta_{4bc}\gamma_{4ca}$	θ_{6abc}
$A, B, C = 0, 1, 1$	$\alpha_{2ab}\beta_{4bc}\gamma_{3ca}$	θ_{7abc}
$A, B, C = 0, 0, 1$	$\alpha_{1ab}\beta_{2bc}\gamma_{3ca}$	θ_{8abc}

Also note that the pairwise factors themselves are not probabilities, and have no constraints on them. It is only when the joint is computed from them that their degrees of freedom are limited.

- (b) (2 points) Consider a MN comprising among all other factors, two specific factors $\phi_1(A, B)$ and $\phi_2(B, C)$. Will changing these factor parameterizations as below change the joint distribution? Justify your answer.

$$\begin{aligned}\phi_1(A, B = b) &:= \phi_1(A, B = b) \lambda_b \\ \phi_2(B = b, C) &:= \phi_2(B = b, C) (1/\lambda_b),\end{aligned}$$

where λ_1 and λ_2 are any positive constants.

Solution: Solution: Consider the joint distribution over the MN:

$$P(A, B, C, W) = \frac{1}{Z} \phi(A, B) \phi(B, C) \phi(A, B, C, W) \quad (1a)$$

Where $\phi(A, B, C, W)$ denotes all other factors over the variables.

The partition function Z can be found as follows:

$$Z = \sum_{A,B,C,W} \phi(A,B)\phi(B,C)\phi(A,B,C,W) \quad (1b)$$

Replacing $\phi(A, B = b)$ and $\phi(B = b, C)$ in 1a and 1b, we get,

$$Z = \sum_{A,C,W} \sum_{B=b} \phi(A, B = b)\lambda_b \phi(B = b, C) \frac{1}{\lambda_b} \phi(A, B, C, W) \quad (1c)$$

$$= \sum_{A,B,C,W} \sum_{B=b} \phi(A, B)\phi(B, C)\phi(A, B, C, W) \quad (1d)$$

and,

$$P(A, B = b, C, W) = \frac{1}{Z} \phi(A, B = b)\lambda_b \phi(B = b, C) \frac{1}{\lambda_b} \phi(A, B, C, W) \quad (1e)$$

$$\therefore P(A, B, C, W) = \frac{1}{Z} \phi(A, B)\phi(B, C)\phi(A, B, C, W) \quad (1f)$$

Thus, from equation 1a and 1f, the joint distribution does not change.

- (c) (3 points) Consider the following factorisation of a distribution P over 4 r.v.s that take values in $\{1, 2, \dots, a\}$

$$P(X_1 = x_1, X_2 = x_2, X_3 = x_3, X_4 = x_4) = \frac{1}{Z} \psi_1(x_1, x_2) \psi_2(x_2, x_3) \psi_3(x_3, x_4) \psi_4(x_1, x_4) \psi_5(x_2, x_4)$$

Let $\mathcal{H}$ be the Markov network for the factorisation above. If Q is a distribution that factorizes according to $\mathcal{H}$, what can be said about the form of the distribution Q ? How many parameters would be required to represent such a distribution Q ? Compare this with the number of parameters required to represent the distribution P . (You may assume a is large enough that $a - 1 \approx a$ and a^k is negligible when compared to a^{k+1} .)

Solution: Solution:

A general distribution over k variables will always take $a^k - 1$ parameters to represent, irrespective of the graph that it represents.

The form of the distribution, however, will change when it is factorized over maximal cliques. Here, maximal cliques will be A_1, A_2, A_4 and A_2, A_3, A_4 .

The distribution Q will then be factorized as: $Q(X) = \frac{1}{Z} \phi(A_1, A_2, A_3) \phi(A_2, A_3, A_4)$

The number of parameters in this distribution is $2a^3$. For the distribution P , the number of parameters is $5a^2$. Neither of these factorizations have enough parameters to represent the distribution, but the number of parameters in Q greatly exceeds the number of parameters in P (According to the given assumptions).

4. (9 points) [I-EQUIVALENCE AND POSITIVE FRIENDS]

(a) (2 points) How many graphs are I-equivalent to the simple directed chain $X_1 \rightarrow X_2 \cdots \rightarrow X_n$?

Solution: Consider the undirected skeleton $X_1 - X_2 \cdots - X_n$.

Since the simple directed chain contains no v-structures, no graph G' which is I-equivalent to G can contain a v-structure.

Thus, we need to find BNs G' which arise from the undirected skeleton $X_1 - X_2 \cdots - X_n$ but have no v-structures.

We find that there are $n - 1$ such graphs - each created by flipping all edges before index 1 to n .

(b) (2 points) Show a BN G (if any) that is I-equivalent to the Diamond Graph MN H (Student Misconception Example with $I(H) = \{(A \perp C \mid B, D), (B \perp D \mid A, C)\}$), when extra variables are allowed in the BN. That is, you are allowed a few extra variables W in the BN G that are always unobserved s.t. the set of dsep-implied CIs in G over the X variables alone (which holds in the marginal distbn. $P(X) = \sum_w P(X, W)$) is identical to $I(H)$.

Solution: Solution: We need to generate a perfect I-map for the given independencies $I(H)$:

$$I(H) = \{(A \perp C \mid B, D), (B \perp D \mid A, C)\} \quad (2)$$

We prove that no such Bayesian Network exists for $\mathcal{X} = (A, B, C, D, W)$, where W is a set of extra variables.

To find the perfect map, we start with the undirected complete graph over $\mathcal{X}$.

We eliminate the edges $A - C$ and $B - D$ from the complete skeleton to satisfy the independencies in 2.

Note that we cannot remove any more edges between (A, B, C, D) , since this would imply additional independencies, which are not in $I(H)$.

Thus, the undirected trail $A - B - C - D$ exists in our skeleton graph. Note that removing edges between W and (A, B, C, D) is possible but irrelevant to our subsequent arguments, since the undirected trail $A - B - C - D$ would still exist in the skeleton. We need to convert this to a BN, i.e. give directionality to the undirected edges.

Since the trail $A - B - C - D$ exists in the Bayesian Network, there exists at least one v-structure in this trail to prevent a cycle. This v-structure implies additional independencies which are not in $I(H)$. For example, if the BN contains $A \rightarrow B \leftarrow C \leftarrow D \leftarrow A$, then this implies $(A \perp C \mid D)$, since the path $A \rightarrow B \leftarrow C$ is not active until B is observed.

This implies that no BN can be created from the constructed undirected skeleton which maintains the existing independencies but does not add additional independencies.

Thus, a perfect map cannot be constructed which exactly satisfies the given independencies even on adding extra nodes.

TL;DR: Such a graph cannot exist, addition of extra nodes makes no difference, since there will always be an immorality in the extended loop $A \dots B \dots C \dots D \dots A$, implying additional independencies between these variables not existing in the original distribution.

- (c) (2 points) Let $\mathcal{P}$ be a positive distribution, then prove that the Markov Blanket $MB_{\mathcal{P}}(X)$ is unique. [Hint: What happens if two different sets U_1 and U_2 are each minimal $MB_{\mathcal{P}}(X)$?]

Solution: Solution: Consider two different sets U_1 and U_2 which are both Markov Blankets of X in a distribution $\mathcal{P}$.

Then, U_1 and U_2 are both minimal sets such that

$$(X \perp \mathcal{X} - X - U_i \mid U_i) \in I(\mathcal{P}) \quad (3)$$

Let Y be $\mathcal{X} - X - U_1 \cup U_2$.

Since U_1 and U_2 are different, they must contain at least one node not belonging to both, i.e.

$$\exists V \text{ s.t. } V \notin U_1 \cap U_2$$

Without loss of generality, let $V \in U_1$.
 $\therefore V \notin U_2$.

Then, since U_1 is a minimal $MB_P(X)$, removal of any node from U_1 would create an active trail between X and Y , and would thus not satisfy equation 3.

Thus, there exists an active trail $X \rightleftharpoons V \rightleftharpoons Y$.

Consider the set U_2 . Since U_2 is a Markov Blanket, U_2 satisfies equation 3. However, since V is not in U_2 , there exists an active trail between X and Y outside U_2 , from the discussion above. This is a contradiction.

Thus, there cannot be any node V s.t. $V \in U_1, V \notin U_2$.

By Symmetry, there cannot be any node V s.t. $V \in U_2, V \notin U_1$.

$\therefore U_1 = U_2$

Thus, the Markov blanket is unique.

- (d) (3 points) If a DAG G has no immoralities, then prove that G is chordal (i.e, the underlying undirected graph is chordal). Does such DAG G always have a perfect MN I-map?

Solution: Solution: A Chordal Graph is one where every loop for $k \geq 4$ has a chord.

Proof: Case I: The DAG has no loops of size $k \geq 4$. Then it is trivially chordal.

Case II: The graph has a loop of size $k \geq 4$. We show the proof for $k = 4$. The same logic can be extended for a loop of any size.

Consider (A, B, C, D) to be the 4 nodes in the loop.

Since the graph is a DAG, there cannot be a cycle $A \rightarrow B \rightarrow C \rightarrow D$ or the opposite. Thus, at least one arrow is inverted, i.e. at least one v-structure exists in the graph.

Without loss of generality, assume that $A \rightarrow B \leftarrow C$ is the v-structure. Then, since no immoralities exist in the graph, the v-structure is covered, i.e. either $A \rightarrow C$ or $A \leftarrow C$ exists

in the graph.

Then, the underlying graph for the loop $A - B - C - D$ is chordal, since $A - C$ is a chord.

Such an argument can be made for every loop in the graph for size $\geq 4 \implies$ The graph is chordal.

- (e) (3 points) [BONUS] Solve Exercise 4.1 from [KF] book related to the non-positive distribution example in Page 116. It will show how Hammersley-Clifford theorem doesn't hold for this distribution.
5. (8 points) [GAUSS EYES MARKOV] A multivariate normal distribution can be viewed either as a BN or as a MN as pointed in our Friday reading. Here we will motivate the latter. Let Σ be the covariance matrix of a set of p variables X . Consider the partial covariance matrix $\Sigma_{a|b} := \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$ between the two subsets of variables $X_a = (X_1, X_2)$ consisting of the first two, and X_b the rest. This is the covariance matrix between these two variables, after linear adjustment for all the rest. In the Gaussian distribution, this is the covariance matrix of the conditional distribution of $X_a|X_b$, which is also multivariate Gaussian. Define $\Theta = \Sigma^{-1}$.
- (a) (2 points) Show that $\Sigma_{a|b} = (\Theta_{aa})^{-1}$.

Solution: Σ is a $p \times p$ covariance matrix.

$$\Sigma = \begin{bmatrix} \Sigma_{aa_{2 \times 2}} & \Sigma_{ab_{2 \times (p-2)}} \\ \Sigma_{ba_{(p-2) \times 2}} & \Sigma_{bb_{(p-2) \times (p-2)}} \end{bmatrix}$$

$$\Theta = \Sigma^{-1} = \begin{bmatrix} \Theta_{aa_{2 \times 2}} & \Theta_{ab_{2 \times (p-2)}} \\ \Theta_{ba_{(p-2) \times 2}} & \Theta_{bb_{(p-2) \times (p-2)}} \end{bmatrix}$$

$\Sigma\Theta = I_{p \times p}$. Therefore,

$$\Sigma_{aa}\Theta_{aa} + \Sigma_{ab}\Theta_{ba} = I_{2 \times 2} \quad (4a)$$

$$\Sigma_{aa}\Theta_{ab} + \Sigma_{ab}\Theta_{bb} = 0_{2 \times (p-2)} \quad (4b)$$

$$\Sigma_{ba}\Theta_{aa} + \Sigma_{bb}\Theta_{ba} = 0_{(p-2) \times 2} \quad (4c)$$

$$\Sigma_{ba}\Theta_{ab} + \Sigma_{bb}\Theta_{bb} = I_{(p-2) \times (p-2)} \quad (4d)$$

From Equation 4c, $\Theta_{ba} = -\Sigma_{bb}^{-1}\Sigma_{ba}\Theta_{aa}$

Substituting this in Equation 4a, we get:

$$\Sigma_{aa}\Theta_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}\Theta_{aa} = I_{2 \times 2}$$

$$\therefore \Theta_{aa} (\Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}) = I$$

$$\implies \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba} = \Theta_{aa}^{-1}$$

$$\implies \Sigma_{a|b} = \Theta_{aa}^{-1}$$

- (b) (3 points) Show that if any off-diagonal element of Θ is zero, then the partial covariance between the corresponding variables is zero. Does this suggest a (minimal I-map) MN H for this distribution, with an edge between any variable pair exhibiting non-zero partial covariance? Why or why not?
- (c) (1 point) Can any multivariate joint distbn. in the exponential family be expressed as a MN distbn. P? If so, specify the (local) factors in this Markov Network MN and their scopes (using terms in the exponential family defn. given in the last assignment).
- (d) (2 points) Express a multivariate normal distribution as an exponential family in terms of its natural parameters: $\Theta = \Sigma^{-1}$ and $\eta := \Sigma^{-1}\mu$, and thereby show that the MN for this exponential family is identical to the MN derived in part (b) above.

Solution: $P(X | \Theta, \eta) = \exp\{-X^T \Theta X - \eta^T X + c\}$

6. (7 points) [MINIMAL BN I-MAPS] Solve Exercise 3.11 from [KF] book (first two parts and the additional part below).
- (a) (3 points) Give the minimal I-map for the marginal distribution over all r.v.s except *Alarm* in the Burglary Alarm Network.

Solution:

We need to build a minimal I-map that satisfies most independencies but also captures *all* dependencies. The independencies in the original network not containing alarm are:
 $(B \perp E, N, TV), (E \perp N, TV), (TV \perp N, M), (N \mid J)$

The dependencies are:

$(B \not\perp E \mid A), (B \not\perp N \mid M), (B \not\perp TV \mid J), (E \not\perp N \mid M), (E \not\perp TV \mid J), (TV \not\perp M \mid J), (N \not\perp J \mid M), (TV \not\perp N \mid J, M)$

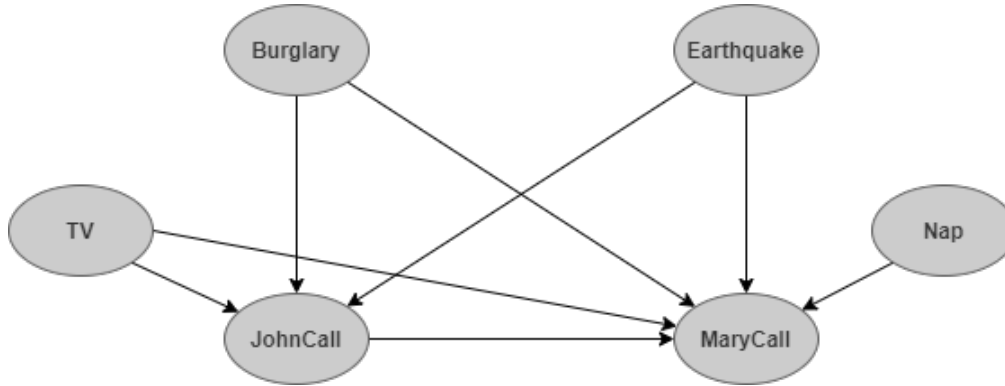


Figure 1: Figure 6: Minimal I-Map retrieved from a variable ordering

Maintaining the dependencies in the original graph as well as a maximum number of independencies, we get the graph in Figure 1.

Note that the given minimal I-map satisfies all the dependencies in the original distribution.

- (b) (3 points) Give an algorithm that outputs a minimal I-map G' as in part (a), but when removing (marginalizing out) a r.v. in a general DAG G .

(Hint: Recall that the minimal BN I-map is not always unique and depends on the node ordering; employ a node ordering that simplifies your algorithm by adding as few edges as possible to G to obtain G').

Solution: The following algorithm can be used to generate a minimal I-map G' after marginalizing out a r.v. A from a graph G :

Procedure Build-Minimal-Marginalized-I-Map (
 $X_1, \dots, X_n$ // an ordering of random variables
 G // an original graph
 A // a variable to be marginalized from G
)

- For $X_p \in \text{Pa}(A), X_c \in \text{Children}(A)$, add edge $X_p \rightarrow X_c$
- For $X_C, X_c \in \text{Children}(A)$, add an edge $X_C \rightarrow X_c$, making sure no cycles are created in the graph. (The proof that this is always possible is not shown here).
- For $X_C, X_c \in \text{Children}(A)$ such that $X_C \rightarrow X_c \in G'$, add an edge $X_{p'} \rightarrow X_c$, where $X_{p'}$ are Parents of X_C .

The third step is performed to prevent the formation of a new independency $(X_c \perp X_{p'} \mid \text{Pa}_G(X_c))$, where, X_c, X_C are children of A, the edge $X_C \rightarrow X_c$ exists in the graph G' , and $X_{p'}$ is a parent of X_C . For example, in the burglar alarm example above, without the edge $TV \rightarrow \text{MaryCall}$ in G' , the independency $(TV \perp \text{MaryCall} \mid \text{JohnCall}, \text{Burglary}, \text{Earthquake})$ is created in G' which is not present in the original graph due to the active trail $TV \rightarrow \text{JohnCall} \leftarrow \text{Alarm} \rightarrow \text{MaryCall}$.

- (c) (1 point) What happens if your algorithm in part (b) is applied to remove the class variable Y in the Naive Bayes model $P(Y, X) = P(Y) \prod_{i=1}^P P(X_i|Y)$?

Solution: Then, an edge is added between every pair of nodes X_i and X_j such that no cycles are created. For example, this can be done by adding unidirectional edges between each pair of nodes (i.e. all edges have same directionality).